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Appendix A. Fine-tuning hyperparameters

Table 6, 7, 8, and 9 present the hyperparameter values of the BERT, RoBERTa, XLNet, and ELMo models, respectively, in the fine-tuning experiments on different NLP tasks. Those hyperparameters not included in the tables were used with their default values specified by the original models.

Hyperparameter	TC	SA	QA	NER	SS
Max sequence length	64	64	256	64	64
Batch size	12	12	2	12	12
Learning rate	2e-5	1e-5	3e-5	1e-5	2e-5
Num epochs	20	15	10	20	20

Table 6: Fine-tuning hyperparameters for the BERT language model on different tasks. TC: Text Classification, SA: Sentiment Analysis, QA: Question Answering, NER: Named Entity Recognition, SS: Semantic Similarity.

Hyperparameter	TC	SA	QA	NER	SS
Max sequence length	64	64	256	64	64
Batch size	12	12	2	12	12
Learning rate	1e-5	1e-5	1.5e-5	2e-5	2e-5
Weight decay	0.1	0.1	0.01	0.1	0.01
Learning rate decay	Linear	Linear	Linear	Linear	Linear
Warmup ratio	0.06	0.06	0.06	0.05	0.05
Num epochs	20	15	10	20	20

Table 7: Fine-tuning hyperparameters for the RoBERTa language model on different tasks. TC: Text Classification, SA: Sentiment Analysis, QA: Question Answering, NER: Named Entity Recognition, SS: Semantic Similarity.

Hyperparameter	TC	SA	QA	NER	SS
Max sequence length	128	128	256	128	128
Batch size	8	8	2	8	8
Learning rate	2e-5	2e-5	2.e-5	1e-5	1e-5
Num steps	6K	6K	4K	6K	6K
Learning rate decay	Linear	Linear	Linear	Linear	Linear

Table 8: Fine-tuning hyperparameters for the XLNet language model on different tasks. TC: Text Classification, SA: Sentiment Analysis, QA: Question Answering, NER: Named Entity Recognition, SS: Semantic Similarity.

Hyperparameter	TC	SA	QA	NER	SS
n_highway	2	2	2	2	2
Droupout	0.2	0.2	0.2	0.2	0.2
Batch size	128	128	256	128	128
Projection dim	512	512	512	512	512
Num epochs	20	20	10	20	20

Table 9: Fine-tuning hyperparameters for the ELMo language model on different tasks. TC: Text Classification, SA: Sentiment Analysis, QA: Question Answering, NER: Named Entity Recognition, SS: Semantic Similarity.